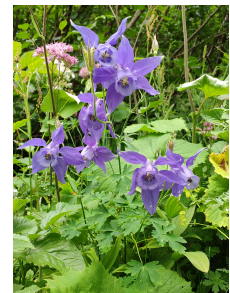


GNN-SDM: Species Distribution Modelling with Graph Neural Networks for Swiss Flora at 30 m Resolution

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From left: *Cypripedium calceolus*, *Potentilla erecta*, *Aquilegia alpina*. Photos: Info Flora, CC BY-NC-SA 4.0.

Traditional species distribution models treat occurrences as independent points, ignoring the spatial structure of landscapes. We adapt the SOM-GNN-SDM framework (Wu et al., 2025) to Swiss vascular plants at 30 m resolution, constructing a landscape graph of 166,464 patches from 12 environmental features via Self-Organising Map clustering. A GraphSAGE architecture tuned for sessile flora models habitat suitability for 2,696 species, outperforming a Random Forest baseline for 85.3 % of species (mean AUC 0.912 vs. 0.874). Circuit-theory connectivity analysis across 241 nationally threatened species reveals that 41 % of high-priority habitat—patches combining high suitability with critical corridor function—currently lacks formal protection. The pipeline identifies spatially explicit conservation gaps.

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1. Introduction

Biodiversity is declining at an unprecedented rate worldwide, driven by habitat loss, fragmentation, climate change, and land-use intensification (IPBES, 2019). Switzerland, despite its relatively small area, harbours a unique plant diversity owing to its pronounced altitudinal gradients and mosaic of biogeographic regions. Nevertheless, a third of all assessed species and half of all habitat types are threatened (Swiss Federal Council, 2022), with 28% of vascular plants considered endangered or regionally extinct and an additional 14% near-threatened (Info Flora, 2016). The main drivers, such as soil sealing, landscape fragmentation, intensive land use, and nitrogen inputs, continue to erode habitat quality and connectivity (Swiss Federal Council, 2022; Federal Office for the Environment, 2025). Effective conservation planning requires spatially explicit predictions of where species can persist, and which landscape elements link suitable habitat patches into functional networks. Species distribution models (SDMs) have become a tool in conservation efforts, relating georeferenced occurrence records to environmental predictors to map habitat suitability across space (Elith and Leathwick, 2009; Guisan and Thuiller, 2005). Classical approaches, such as generalised linear models, maximum entropy (Phillips et al., 2006), and ensemble methods such as Random Forests (Breiman, 2001), treat each occurrence as an independent sample drawn from an environmental niche. This point-based paradigm ignores the spatial structure of landscapes: two patches with identical environmental conditions may differ in ecological value depending on their size, shape, and connectivity to neighbouring habitat. For species whose long-term persistence depends on seed dispersal between habitat patches, such an omission can lead to misleading suitability estimates and suboptimal conservation recommendations.

Graph Neural Networks (GNNs) offer a way to incorporate landscape structure into SDMs, by representing the landscape as a graph, where nodes are ecologically homogeneous patches and edges encode spatial adjacency. Wu et al. (2025) introduced the SOM-GNN-SDM framework, demonstrating that a GraphSAGE architecture (Hamilton et al., 2017) trained on landscape graphs outperforms traditional SDMs for virtual and real species across continental extents. Their approach constructs landscape patches via Self-Organising Maps (SOMs) (Kohonen, 1982), builds an adjacency graph, and propagates information across the graph to produce suitability predictions that are spatially coherent. Additionally, the graph structure enables conservation applications beyond suitability mapping. The Swiss federal government has identified the lack of an ecological infrastructure that protects and connects core biodiversity areas as a key deficit, setting a target of 17% of national territory as connected core areas by 2030, up from 13.4% today (Swiss Federal Council, 2022; Federal Office for the Environment, 2025). Because the landscape graph explicitly encodes connectivity, it can be analysed using circuit theory and centrality metrics to identify critical corridors: patches whose protection would maintain or restore habitat connectivity for threatened species.

In this work, we adapt the SOM-GNN-SDM framework to Swiss vascular flora at 30 m resolution, presenting a finer grain than the original continental-scale application. We construct a landscape graph of 166,464 patches from 12 environmental features using SOM-based clustering and connected-component labelling, train per-species GraphSAGE models for 2,696 qualifying plant species, and benchmark against a Random Forest baseline. We further apply circuit-theory connectivity analysis to the resulting suitability maps to identify conservation priority areas and quantify protection gaps relative to the existing protected-area network.

Specifically, we address the following research questions:

2. Data and Study Area

1. Can GNN-SDM outperform a traditional Random Forest at patch-level species distribution modelling for Swiss plants?
2. Does the graph structure confer greater benefit for rare and vulnerable species, whose distributions are more fragmented?
3. What GNN architecture is best suited for flora—given that plants, unlike the mobile fauna studied by Wu et al. (2025), have limited active dispersal?
4. Can the graph structure be used to identify spatial gaps in the protected-area network where additional conservation measures would be most beneficial?

2. Data and Study Area

2.1. Study Area

The study area encompasses the entirety of Switzerland and a surrounding buffer zone, defined by the spatial extent of the CHELSA high-resolution climate dataset (WSL, 2024). The bounding box spans from 45.57° N to 48.07° N latitude and 5.72° E to 10.66° E longitude, covering approximately 41,400 km² of land area. This extent includes a border buffer of roughly 10 km into neighbouring countries to avoid edge effects in spatial analyses such as flow accumulation and neighbourhood-based feature computation.

All environmental layers are aligned to a common 30 m master grid derived from the Copernicus GLO-30 Digital Elevation Model (European Space Agency, 2023), projected in geographic coordinates (EPSG:4326). Switzerland's topography ranges from 193 m (Lake Maggiore) to 4,634 m (Dufourspitze), creating steep environmental gradients over short distances, motivating the use of high-resolution modelling.

2.2. Environmental Features

We compiled 21 candidate environmental features from four thematic groups (Table 1): terrain, land cover, climate, and biological/anthropogenic variables. All layers were processed to the 30 m master grid.

Terrain features. Elevation was obtained directly from the Copernicus GLO-30 DEM (European Space Agency, 2023). Slope and aspect were derived using first-order finite differences at 30 m resolution. Aspect was encoded as two variables (sin and cos of the azimuth angle) to avoid the circular discontinuity at 0°/360°. The Topographic Wetness Index (TWI) was computed using D8 flow direction and accumulation via the `pysheds` library, with pit-filling and depression resolution applied beforehand. Profile and plan curvature were derived from second-order finite differences and smoothed with a 5×5 mean filter to reduce noise.

Land cover. Six land-cover fraction layers were computed from ESA WorldCover 2021 (European Space Agency, 2022) at its native 10 m resolution. For each class, a binary mask was reprojected to the 30 m grid using average resampling, yielding the fraction of each 30 m pixel covered by that land-cover type. The six classes retained for Switzerland are: tree cover, grassland, cropland, built-up, snow/ice, and permanent water bodies.

2. Data and Study Area

Table 1: Environmental predictor variables compiled for the study. All layers are aligned to the 30 m master grid. The 12 features selected for modelling (after feature selection, Section 2.5) are marked with •.

Group	Feature	Source	Native res.	Sel.
Terrain	Elevation	Copernicus GLO-30 DEM	30 m	•
	Slope	Derived from DEM	30 m	•
	Aspect (sin, cos)	Derived from DEM	30 m	•
	Topographic Wetness Index	D8 flow accumulation	30 m	•
	Profile curvature	2 nd -order finite diff.	30 m	
	Plan curvature	2 nd -order finite diff.	30 m	
Land cover	Tree cover fraction	ESA WorldCover 2021	10 m	•
	Grassland fraction	ESA WorldCover 2021	10 m	•
	Cropland fraction	ESA WorldCover 2021	10 m	•
	Built-up fraction	ESA WorldCover 2021	10 m	
	Snow/ice fraction	ESA WorldCover 2021	10 m	
	Water fraction	ESA WorldCover 2021	10 m	
Climate	Bio1 (annual mean temp.)	CHELSACh	25 m	•
	Bio4 (temp. seasonality)	CHELSACh	25 m	•
	Bio12 (annual precip.)	CHELSACh	25 m	•
	Bio15 (precip. seasonality)	CHELSACh	25 m	
	Bio18 (precip. warmest quarter)	CHELSACh	25 m	
Biological	Canopy height	ETH Global Canopy Height	10 m	•
	NDVI (annual max)	GIMMS MODIS	250 m	
	Human Footprint	Mu et al. (2022)	1 km	•
	Forest-edge distance	Derived from tree cover	30 m	

2. Data and Study Area

Climate. Five bioclimatic variables were derived from CHELSA monthly temperature and precipitation normals (1981–2010) at 25 m resolution (WSL, 2024): annual mean temperature (Bio1), temperature seasonality (Bio4, standard deviation), annual precipitation (Bio12), precipitation seasonality (Bio15, coefficient of variation), and precipitation of the warmest quarter (Bio18). Each variable was regridded to the 30 m master grid using bilinear interpolation.

Biological and anthropogenic features. Canopy height was obtained from the ETH Global Canopy Height Model at 10 m resolution (Lang et al., 2023) and regridded to 30 m via bilinear resampling. The annual maximum NDVI was computed from GIMMS MODIS 8-day composites (250 m, growing season DOY 97–273) and regridded to 30 m. The global Human Footprint index (Mu et al., 2022) at 1 km was bilinearly downscaled. Forest-edge distance was computed as a signed Euclidean distance transform from the tree-cover fraction layer (threshold 0.5): positive values indicate distance from forest, negative values indicate depth within forest.

2.3. Species Occurrence Data

Species occurrence records for vascular plants in Switzerland were obtained from the Global Biodiversity Information Facility (GBIF) open data snapshot hosted on Amazon S3 (GBIF.org, 2024). The raw dataset contained approximately 22.7 million records within the study extent. Records were filtered to retain only those with valid geographic coordinates, taxonomic rank at species level, and—critically—a coordinate uncertainty ≤ 1000 m. This precision filter removed 86 % of records, reducing the dataset to 3.14 million high-quality occurrences. The strict threshold ensures that each record can be reliably assigned to the correct 30 m landscape patch; imprecise observations (e.g. municipality centroids) would introduce systematic noise into the presence labels.

After spatial filtering, occurrence coordinates were mapped onto the landscape patch raster (Section 3.1), and each species was characterised by its set of presence patches. For modelling purposes, we retained only species with at least 20 presence patches, yielding **2,696 qualifying species**. This patch-based threshold (rather than a raw record count) directly reflects the spatial information available for training: a species needs geographically distinct observations spread across multiple habitat patches for the model to learn meaningful environmental associations.

2.4. Protected Areas

Protected area boundaries for Switzerland were obtained from the World Database on Protected Areas (WDPA), filtered to nationally designated sites within the study extent. The dataset includes national parks, nature reserves, landscape protection areas, and Ramsar sites. These boundaries are used in the gap analysis (Section 4.5) to quantify how much of the identified priority habitat is already protected.

2.5. Feature Selection

To reduce multicollinearity and computational cost, we performed feature selection using Random Forest importance scores. A patch-level RF classifier was trained for six representative

3. Methods

species spanning different habitat types and prevalence levels. Feature importance was computed as the mean decrease in Gini impurity, averaged across species.

From the initial 21 candidate features, 12 were retained for downstream modelling (Table 1). The excluded features were either highly correlated with retained variables (e.g., Bio15 with Bio12; plan curvature with profile curvature) or contributed negligible predictive power (e.g., NDVI at 250 m resolution added little beyond the 10 m canopy height and tree-cover fraction). The selected feature set provides good predictive coverage without redundancy, and keeps dimensionality low enough for the SOM step.

3. Methods

This section describes the methodological pipeline for constructing a graph-based species distribution model (SOM-GNN-SDM) for Swiss flora at 30 m resolution, following the framework proposed by Wu et al. (2025). The pipeline consists of three stages: landscape patch construction via Self-Organising Map clustering and connected- component analysis, (2) a Random Forest baseline, and (3) the SOM-GNN-SDM itself.

3.1. Landscape Patch Construction

The landscape patch construction transforms the continuous 30 m raster feature stack into a graph representation where nodes correspond to ecologically homogeneous habitat patches and edges encode spatial adjacency.

3.1.1. SOM Clustering

A Self-Organising Map (SOM; Kohonen, 1982) was trained to identify recurring landscape types from the 12 selected environmental features. The SOM was configured as a 64×64 hexagonal, toroidal grid and initialised with PCA. Training used a subsample of 500 000 pixels (drawn from the ~ 160 M valid pixels in the study extent), preprocessed with a `RobustScaler` and domain-specific post-scaling (land-cover fractions up-weighted $3\times$; forest-edge distance clipped to $[-5, 5]$).

The SOM was trained for 150 epochs with a neighbourhood radius decaying from 64 to 16 and a learning rate decaying from 0.05 to 0.016. Convergence was assessed on a held-out 20% validation set using quantisation error (QE) and topographic error (TE).

After training, the codebook vectors were hierarchically clustered using Ward's linkage, and the dendrogram was cut to yield 30 landscape types. Every valid pixel in the study extent was then assigned to its best-matching unit (BMU) via a FAISS exact- L_2 index, producing a categorical landscape-type raster at 30 m resolution.

3.1.2. Connected Components and Patch Merging

Spatially contiguous regions of the same landscape type were identified using connected-component labelling (`skimage.measure.label`). To avoid an excessive number of tiny fragments, patches smaller than 100 pixels ($\sim 0.09 \text{ km}^2$) were merged into their most similar neighbour using a mapping-table approach. This procedure yielded **166,464 landscape patches** covering the full study extent.

For each patch, the mean value of the 12 selected environmental features was computed from the underlying raster stack, producing a $166,464 \times 12$ node-feature matrix.

3.1.3. Patch Network

An undirected graph was constructed by connecting patches that share at least one boundary pixel. The resulting `NetworkX` graph contains **166,464 nodes** and **465,833 edges**. Node attributes are the mean environmental features; no edge weights were used.

3.2. Baseline: Random Forest

A per-species Random Forest (Breiman, 2001) classifier served as the non-spatial baseline. For each of the 2,696 qualifying species (≥ 20 presence patches), the following procedure was applied:

1. **Presence labels.** GBIF occurrence coordinates were mapped to the patch-label raster, yielding a set of presence patches per species.
2. **Pseudo-absence generation.** Background patches were sampled at a 3:1 ratio (absence : presence) from patches without recorded occurrences, using a fixed random seed for reproducibility.
3. **Model training.** A `scikit-learn` Random Forest was trained on the patch-level mean features using tuned hyperparameters (see below).
4. **Evaluation.** Performance was assessed via 5-fold cross-validated metrics: AUC, accuracy, precision, recall, F1, MCC, Cohen's κ , and TSS.

Hyperparameter tuning was performed via `GridSearchCV` on the same six representative species used for the GNN architecture search, covering diverse habitat types across Switzerland. The best hyperparameters from this limited tuning were then applied uniformly to all 2,696 species.

3.3. GNN-SDM

3.3.1. Model Architecture

The SOM-GNN-SDM is based on GraphSAGE (Hamilton et al., 2017), implemented in PyTorch Geometric (Fey and Lenssen, 2019). At each layer l , a node v aggregates information from

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its neighbourhood $\mathcal{N}(v)$ using the mean aggregator and combines it with its own representation:

$$\mathbf{h}_{\mathcal{N}(v)}^{(l)} = \frac{1}{|\mathcal{N}(v)|} \sum_{u \in \mathcal{N}(v)} \mathbf{h}_u^{(l-1)} \quad (1)$$

$$\mathbf{h}_v^{(l)} = \sigma\left(\mathbf{W}^{(l)} \cdot \text{CONCAT}\left(\mathbf{h}_v^{(l-1)}, \mathbf{h}_{\mathcal{N}(v)}^{(l)}\right)\right) \quad (2)$$

where $\mathbf{h}_v^{(0)} = \mathbf{x}_v$ is the input feature vector of node v , $\mathbf{W}^{(l)}$ is a learnable weight matrix, and σ is the LeakyReLU activation function. Dropout ($p = 0.2$) is applied after each layer during training. The final node embeddings are passed through a linear layer with sigmoid activation to produce a habitat-suitability score $\hat{y}_v \in [0, 1]$ for every patch:

$$\hat{y}_v = \text{sigmoid}\left(\mathbf{w}^\top \mathbf{h}_v^{(L)} + b\right) \quad (3)$$

where L is the number of GraphSAGE layers.

An architecture search was conducted on six representative species spanning diverse habitats (*Picea abies*, *Fagus sylvatica*, *Trifolium pratense*, *Potentilla erecta*, *Phragmites australis*, *Senecio inaequidens*). Six architectures were compared:

- 1-layer: [16] (417 parameters)
- 2-layer: [24, 12] (1,201 parameters)
- Paper default: [24, 18, 8] (1,787 parameters)
- Wider: [48, 32, 16] (5,361 parameters)
- Extra-wide: [64, 48, 32] (10,929 parameters)
- 4-layer: [32, 24, 16, 8] (3,417 parameters)

The extra-wide 3-layer architecture [64, 48, 32] consistently achieved the highest mean AUC across all test species and was selected for production training. This is notably wider than the [24, 18, 8] architecture proposed by Wu et al. (2025) for virtual fauna species, suggesting that flora—with their sessile nature and finer-grained habitat associations—benefit from greater representational capacity per layer rather than additional depth.

3.3.2. Training Procedure

For each species, the GNN-SDM was trained independently using the following pipeline:

1. **Background selection via random pseudo-absence.** Non-presence patches were sampled at a 3:1 ratio (background : presence) using a fixed random seed. An initial design based on One-Class SVM background selection (as in Wu et al., 2025) was evaluated during development but abandoned in favour of random pseudo-absence for the production run. The random strategy ensures a fair comparison with the RF baseline (which uses same pseudo-absence protocol) and proved more robust across the full species set, avoiding the computational overhead and occasional instability of fitting a kernel SVM on high-dimensional patch features for each of 2,696 species.

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2. **PageRank weighting.** Presence patches were weighted by their PageRank centrality in the landscape graph, following Wu et al. (2025). This assigns higher importance to structurally well-connected patches, reflecting their greater ecological significance as habitat corridors or core areas. Background patches received uniform weight of 1.0.
3. **Train/validation split.** The labelled patches (presence + background) were shuffled and split 80/20 into training and validation sets.
4. **Optimisation.** The model was trained with the Adam optimiser (learning rate = 0.001) using a weighted mean-squared-error loss:

$$\mathcal{L} = \frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} w_i (\hat{y}_i - y_i)^2$$

where $\mathcal{T}$ is the training set, w_i the PageRank-derived weight, $\hat{y}_i$ the predicted suitability, and $y_i \in \{0, 1\}$ the label.

5. **Early stopping.** Validation AUC was evaluated every 10 epochs. Training was halted if no improvement occurred for 50 consecutive epochs (patience). The model state with the best validation AUC was restored for prediction. Maximum training duration was 500 epochs.
6. **Prediction.** The best model was used to predict habitat suitability for all 166,464 patches, yielding a continuous suitability surface per species.

3.3.3. Evaluation

To evaluate the GNN-SDM on the same footing as the RF baseline, a post-hoc evaluation was performed for each species: presence patches and a 3:1 random sample of non-presence patches were combined, and the predicted suitability scores were evaluated using the same eight metrics (AUC, accuracy, precision, recall, F1, MCC, Cohen’s κ , TSS) with a classification threshold of 0.5.

3.3.4. Computational Setup

Production training of all 2,696 species was executed as an AWS SageMaker Training Job on a GPU instance (NVIDIA GPU, CUDA-accelerated). The job supported checkpointing for spot-instance interruptions and completed in approximately 5.7 hours of GPU time (mean 7.6 s per species, 231 epochs on average). Node features were standardised (zero mean, unit variance) before training, and the full graph (166,464 nodes, 931,666 directed edges) was loaded onto the GPU for efficient message passing.

3.4. Conservation Prioritisation

The final stage of the pipeline combines the GNN-SDM habitat suitability predictions with circuit-theory connectivity analysis to produce a multi-species conservation priority surface, which is then overlaid with existing protected areas to identify protection gaps.

3.4.1. Complementarity of Suitability and Connectivity

One might ask whether the circuit-theory step is redundant given that GraphSAGE already incorporates neighbourhood information during message passing. The two components are, however, fundamentally complementary:

- **GNN suitability** is a *local* property. GraphSAGE aggregates features from the 1–2 hop neighbourhood to predict whether a given patch is suitable habitat for a species. The output answers: “Is this place good habitat?”
- **Circuit-theory current flow** is a *global network* property. It treats the entire landscape graph as an electrical circuit and computes which patches act as irreplaceable bridges between isolated population clusters. The output answers: “Is this place critical for maintaining connectivity between populations?”

These two perspectives are orthogonal. A patch may have moderate suitability (GNN score ~ 0.3) yet carry very high current because it lies in the only traversable corridor between two populations. Conversely, a patch in the core of a large population may have high suitability but low current because alternative paths exist. The highest conservation value lies a patches that score highly on both dimensions.

3.4.2. Circuit-Theory Connectivity

For each species, the landscape graph is treated as a resistor network where the conductance of each edge (u, v) is defined as the mean suitability of the two adjacent patches:

$$g_{uv} = \max\left(10^{-4}, \frac{\hat{y}_u + \hat{y}_v}{2}\right) \quad (4)$$

This ensures that current preferentially flows through high-suitability regions (low resistance), reflecting the ecological principle that organisms move more easily through suitable habitat.

Source and ground nodes are defined by the species’ actual observation patches. These patches are grouped into connected components in the landscape graph; components with ≥ 3 patches are ranked by size, and current flow is solved between the top $k = 6$ component pairs using Kirchhoff’s circuit laws. Specifically, for each source–ground pair, one unit of current is injected uniformly across source nodes and extracted uniformly at ground nodes. The resulting system of linear equations (conductance Laplacian) is solved for node voltages using a sparse direct solver (`scipy.sparse.linalg.spsolve`), and the absolute current through each node is accumulated across all pairs.

Species with fewer than 2 qualifying components are excluded, as connectivity cannot be meaningfully assessed.

3.4.3. Species Selection and Filtering

The gap analysis targets nationally threatened species according to the Swiss Red Lists of vascular plants (Info Flora, 2016) and bryophytes (Swissbryophytes, 2022), published by Info Flora on behalf of BAFU (Federal Office for the Environment). These lists apply IUCN criteria at the *national* level, as a species can be globally Least Concern yet nationally Critically Endangered in Switzerland.

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Of the 2,696 GNN-trained species, 489 carry a nationally threatened status (CR, EN, VU, or NT). Two additional filters are applied:

1. **Minimum spatial structure.** Only species with ≥ 50 presence patches are retained, ensuring sufficient spatial extent for meaningful circuit-theory analysis. This yields **285 species** (CR = 2, EN = 20, VU = 93, NT = 170).
2. **Minimum connectivity.** During computation, species whose observation patches form fewer than 2 connected components of size ≥ 3 are excluded, as the circuit-theory framework requires at least one source-ground pair. This removes a further 44 species, leaving 241 species with non-zero current flow.

3.4.4. Normalisation and Aggregation

Per-species current flow values span vastly different scales depending on graph structure and population fragmentation. To ensure each species contributes on a comparable scale to the multi-species aggregation, each species' current vector is mean-normalised:

$$\tilde{c}_v = \frac{c_v}{\bar{c}} \quad (5)$$

where $\bar{c} = \frac{1}{N} \sum_v c_v$ is the species-level mean current. Values are thus expressed as multiples of the species mean: $\tilde{c}_v = 1$ indicates average flow, while $\tilde{c}_v \gg 1$ identifies critical pinch-points. This allows comparison across species whose absolute current magnitudes differ: a species with many observed patches generates higher raw current than one with few observations, yet both may contain equally important bottlenecks. Mean normalisation preserves the relative importance of pinch-points within each species while placing all species on a common scale. Species whose mean current is effectively zero ($\bar{c} < 10^{-12}$) are excluded, leaving 241 of 285 species with non-zero flow.

Species are weighted by national Red List category (CR = 4, EN = 3, VU = 2, NT = 1) to reflect conservation urgency. The multi-species priority surface is then computed as:

$$P_v = \frac{1}{2} \left(\frac{\sum_s w_s \hat{y}_{v,s}}{\sum_s w_s} + \frac{\sum_s w_s \tilde{c}_{v,s}}{\sum_s w_s} \right) \quad (6)$$

where w_s is the threat-category weight of species s , $\hat{y}_{v,s}$ the GNN suitability, and $\tilde{c}_{v,s}$ the normalised current flow at patch v .

3.4.5. Protected Area Overlay

The Swiss protected area network (WDPA; 10,522 polygons) is rasterised onto the 30 m patch grid. Each patch is assigned the fraction of its constituent pixels that fall within any protected area polygon. The analysis is restricted to patches within Swiss territory (majority of pixels inside the national boundary) to avoid false negatives from neighbouring countries.

Conservation gaps are quantified by examining what fraction of high-priority patches (top 5% and top 10% of the priority surface) currently lack any protected-area coverage. These unprotected high-priority patches represent the most urgent candidates for protection expansion.

4. Results and Discussion

4.1. Landscape Graph Construction

The SOM converged after approximately 150 training epochs on the 400 000-pixel subsample (80 % of 500 000, with 20 % held out for validation). Figure 1 shows the evolution of validation quantisation error (QE) and topographic error (TE) across training stages. QE decreases monotonically throughout training, but the codebook feature maps (Figure 10) reveal that this continued improvement comes at a cost: at 150 epochs, the codebook displays smooth gradients with distinct regional structure, whereas at 200 epochs the features fragment into noise as the shrinking neighbourhood radius causes individual neurons to overfit local samples. While QE continues to decrease beyond 150 epochs, the topographic error increases and the codebook loses the spatial coherence needed for meaningful landscape-type clustering. The final validation metrics (QE = 1.54, TE = 0.14) at 150 epochs mark the best trade-off between quantisation fidelity and topological preservation.

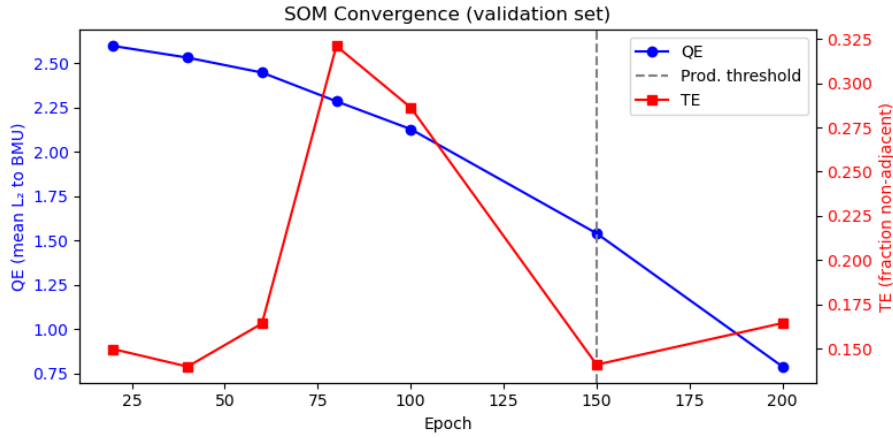


Figure 1: SOM convergence on the held-out validation set. Left axis: quantisation error (mean L_2 distance to best-matching unit). Right axis: topographic error (fraction of samples whose two closest BMUs are non-adjacent on the grid). The dashed vertical line marks the 150-epoch production threshold.

Hierarchical clustering of the 4096 codebook vectors (Ward’s linkage, cut at 30 clusters) partitioned the SOM grid into 30 landscape types representing recurring environmental configurations across Switzerland. Connected-component labelling and merging of fragments smaller than 100 pixels produced the final landscape graph of **166 464 patches** and **465 833 edges**.

The patch size distribution (Figure 9) shows a median of 340 pixels ($\sim 0.31 \text{ km}^2$), with a long tail towards larger contiguous patches in homogeneous terrain (e.g. alpine grassland, dense forest). The minimum enforced size of 100 pixels ($\sim 0.09 \text{ km}^2$) ensures that each patch is large enough to carry reliable feature averages.

The full-extent PCA-coloured patch map (Figure 8) reveals coherent biogeographic structure: similar colours emerge for the Swiss Plateau and the Po Valley south of the Alps, reflecting analogous lowland environmental profiles. High-alpine zones form a distinct cluster, while the southern Pre-Alps and Ticino group with the Jura arc, both of which are characterised by their southward slope and intermediate elevations.

Figure 2 zooms into a $\sim 16 \times 16 \text{ km}$ region north of Bern. Urban patches (Bern city centre) are clearly delineated. The Dählhölzli forest along the Aare, the Schärmenwald near Ittigen, and

4. Results and Discussion

the Tiefenau peninsula emerge as distinct forest-type patches. These structures match local knowledge of the area well.

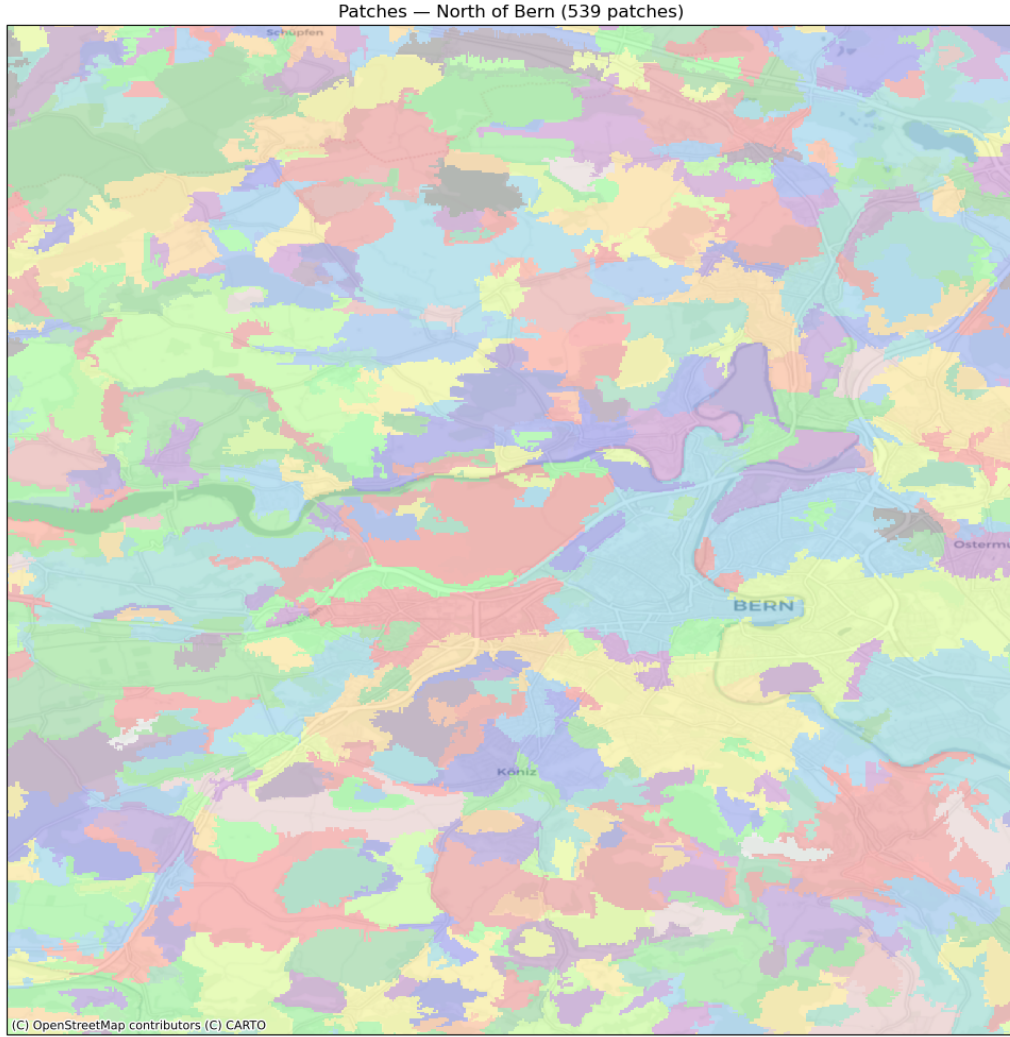


Figure 2: Landscape-type map for the Bern region. Colours represent the 30 landscape types derived from hierarchical clustering of the SOM codebook. The tessellation captures transitions between forested hillsides, agricultural plains, and built-up areas at 30 m resolution.

The SOM-based discretisation serves a dual purpose: it reduces the ~ 160 million valid pixels to a tractable graph of 166 464 nodes while ensuring that patch boundaries follow ecological gradients rather than an arbitrary regular grid. This adaptive resolution means that patches are smaller in heterogeneous landscapes (e.g. valley floors with interspersed land cover) and larger in homogeneous terrain (e.g. high-alpine meadows), mirroring the spatial structure relevant to species distributions.

A limitation of the adjacency-based edge construction is that connections between patches sharing only a single boundary pixel receive the same weight as connections along extensive shared borders. The graph therefore contains “bridge” edges representing tenuous spatial contact (e.g. two forest patches touching through a one-pixel corridor under a road). This is revisited in Section 5.1.

4.2. Random Forest Baseline

A per-species Random Forest served as the non-spatial baseline. Across 2 696 species, the RF achieved a mean AUC of 0.874 (median = 0.877), with 99.3 % of species exceeding 0.7 and 91.6 % exceeding 0.8. Full metric summaries are provided in Appendix B. The RF’s key limitation is its independence assumption: each patch is classified without regard to its landscape context, so the model cannot distinguish isolated habitat fragments from patches embedded in connected corridors.

4.3. Architecture Search

Six GraphSAGE architectures were evaluated on six representative species spanning a range of habitats and prevalence levels (Table 2). Performance was measured by validation AUC averaged over the six species.

Table 2: Architecture search results (validation AUC). Each column corresponds to one of the six representative species; the final column reports the mean across species.

Architecture	<i>P. ab.</i>	<i>F. sy.</i>	<i>T. pr.</i>	<i>P. er.</i>	<i>Ph. au.</i>	<i>S. in.</i>	Mean
1-layer [16]	0.730	0.755	0.670	0.700	0.734	0.877	0.744
2-layer [24,12]	0.742	0.785	0.715	0.725	0.746	0.885	0.766
paper [24,18,8]	0.770	0.791	0.726	0.751	0.771	0.901	0.785
wider [48,32,16]	0.791	0.807	0.752	0.766	0.791	0.908	0.803
extra-wide [64,48,32]	0.796	0.817	0.759	0.776	0.794	0.911	0.809
4-layer [32,24,16,8]	0.787	0.796	0.742	0.750	0.776	0.906	0.793

The extra-wide architecture ([64,48,32], 3 layers) yielded the highest mean AUC (0.809) and was selected for all subsequent experiments. Adding a fourth layer did not improve performance, suggesting that three message-passing hops (covering a ~ 3 -patch spatial neighbourhood) capture the relevant landscape context for Swiss plant species. This contrasts with the original Wu et al., 2025 configuration ([24,18,8]), which was optimised for global-scale fauna; wider hidden layers appear necessary to capture the finer environmental gradients that distinguish Swiss vegetation zones at 30 m resolution.

Graph structure benefit. The improvement from one to three GraphSAGE layers quantifies how much each species profits from neighbourhood information propagated through the landscape graph. The largest gains are observed for alpine and subalpine grassland species (notably *Potentilla erecta* (+0.051 AUC) and *Trifolium pratense* (+0.056)) while the habitat specialist *Senecio inaequidens* benefits least (+0.024), as its distinct urban niche is already well captured by local features alone.

Potentilla erecta (alpine grassland) provides a good illustration. This species occupies connected subalpine meadow patches in distinct elevation bands, shaped by climate and topography rather than human management. With a single GraphSAGE layer, predictions rely solely on local patch features; isolated meadow patches receive unreliable scores because the model lacks spatial context. With three layers, the GNN propagates suitability information through adjacent patches, correctly identifying connected alpine grassland belts as suitable habitat.

4.4. SOM-GNN-SDM Performance

The production GNN-SDM was trained for all 2,696 qualifying species using the extra-wide GraphSAGE architecture ([64,48,32]). Training completed in 5.7 hours of GPU time (mean 7.6 s and 231 epochs per species). Early stopping triggered for 90.4 % of species before reaching the 500-epoch maximum (Appendix C).

Discrimination. Across all 2,696 species, the GNN-SDM achieved a mean evaluation AUC of **0.912** (median = 0.923, $\sigma = 0.055$), with 90.4 % of species exceeding 0.85. The validation AUC used for early stopping averaged 0.886, with 2,437 species (90.4 %) above 0.8 and only 14 species below 0.6.

TSS and classification skill. The True Skill Statistic (defined as sensitivity + specificity – 1 and independent of prevalence) provides a threshold-dependent measure of classification accuracy. At the threshold maximising TSS, the GNN-SDM achieved a mean TSS of **0.583** ($\Delta = +0.153$ over RF). This indicates that the model correctly identifies both presence and absence patches well above chance, despite the extreme class imbalance (median species prevalence ~ 0.8 % of patches).

Comparison with RF baseline. Table 3 summarises the head-to-head comparison. The GNN-SDM outperforms the RF on every metric, with the largest gains in recall (+0.137), F1 (+0.126), and TSS (+0.153). On a per-species basis, the GNN achieves higher AUC than the RF for **2,299 of 2,696 species (85.3 %)**.

Table 3: Mean evaluation metrics across 2,696 species for the Random Forest baseline and SOM-GNN-SDM. Δ denotes the GNN improvement.

Metric	RF	GNN	Δ
AUC	0.874	0.912	+0.038
Accuracy	0.832	0.878	+0.046
Precision	0.733	0.820	+0.087
Recall	0.481	0.618	+0.137
F1	0.561	0.688	+0.126
MCC	0.498	0.640	+0.142
TSS	0.430	0.583	+0.153

The largest gains are in recall and TSS. The RF tends toward conservative predictions (high precision, low recall), missing many true presence patches that the GNN recovers through graph propagation.

Species with spatially structured distributions benefit most from neighbourhood aggregation, while widespread generalists show smaller gains because their habitat signal is already well captured by local features alone. The RF remains a strong baseline (AUC = 0.874): the 12 selected features carry strong predictive power even without spatial context. The GNN primarily improves predictions at patch boundaries and in fragmented landscapes.

Potentilla erecta (Figure 4) illustrates the difference: the GNN produces sharp habitat boundaries along connected alpine grassland belts, while the RF assigns diffuse moderate scores across the entire elevation band.

4. Results and Discussion

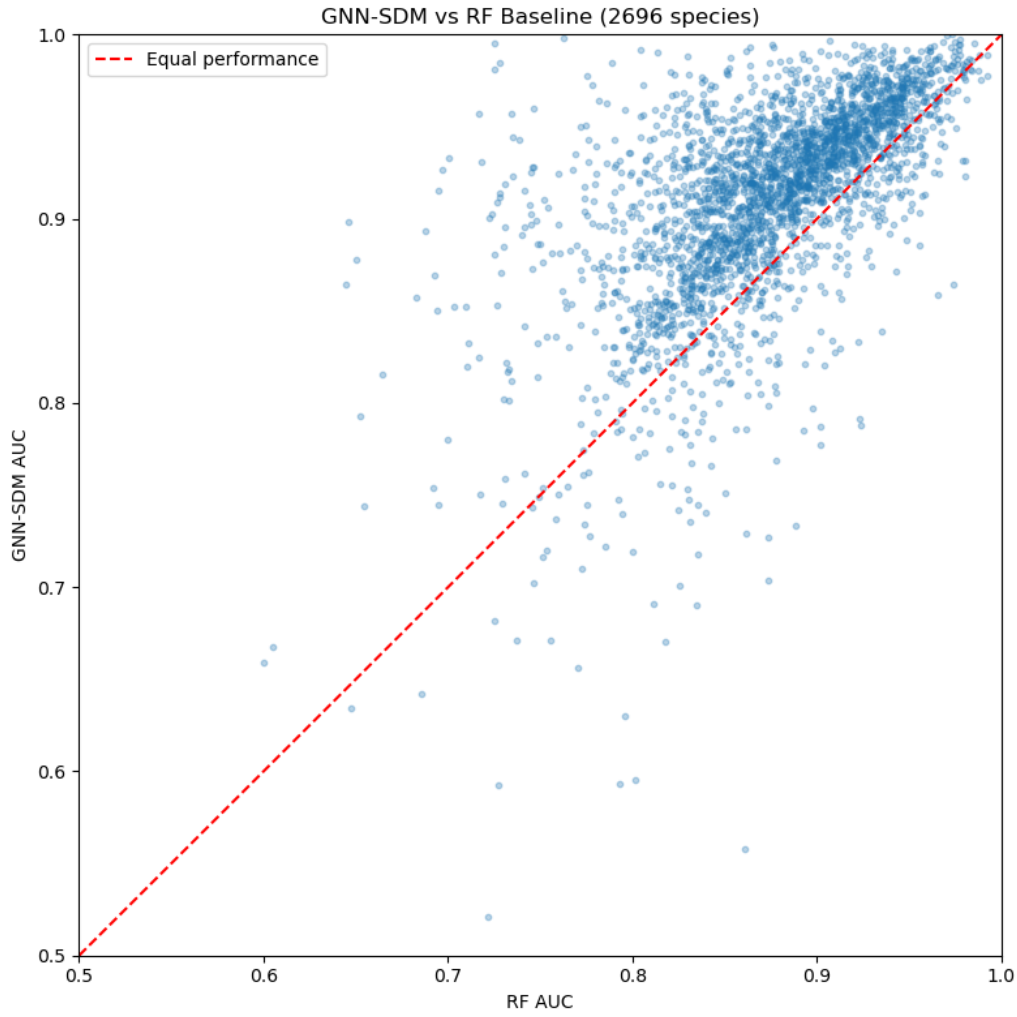


Figure 3: Per-species AUC comparison between SOM-GNN-SDM and Random Forest baseline across 2,696 species. Points above the diagonal indicate species where the GNN outperforms the RF (85.3 % of species). Colour indicates observation prevalence.

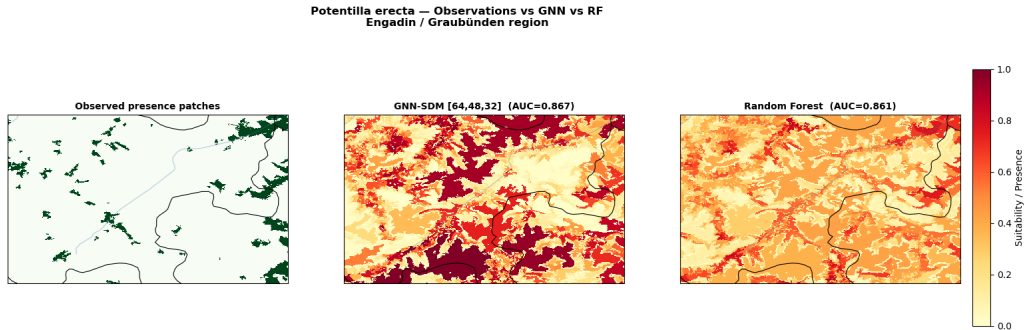


Figure 4: Habitat suitability map for *Potentilla erecta* comparing RF (left) and GNN-SDM (right). The GNN produces sharper habitat delineation along connected alpine grassland belts, leveraging graph structure to resolve ambiguous patches at forest–meadow transitions.

4. Results and Discussion

Common vs. threatened species. A detailed evaluation on 12 selected species (6 common, 6 vulnerable) shows no systematic performance gap: vulnerable species achieve a mean AUC of 0.874 and TSS of 0.708, compared to 0.863 and 0.604 for common species (Figure 5). The slightly higher scores for vulnerable species reflect their narrower ecological niches, which are easier to discriminate from the landscape background.

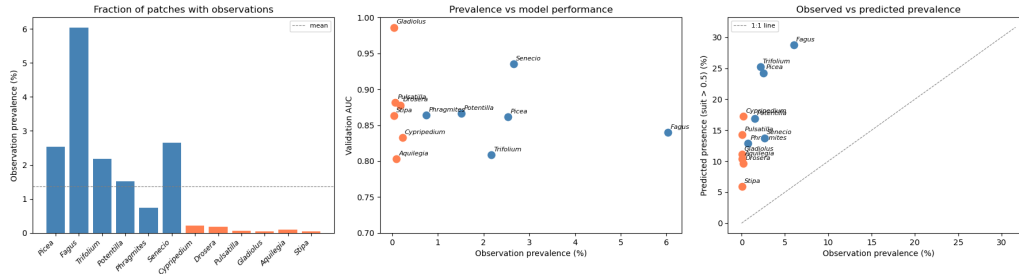


Figure 5: GNN-SDM performance for 6 common vs. 6 vulnerable species. Vulnerable species achieve comparable or higher AUC and TSS despite having far fewer observation patches, confirming the model generalises to rare species.

4.5. Conservation Prioritisation

The circuit-theory connectivity analysis was applied to 285 nationally threatened species (Swiss Red List categories CR, EN, VU, NT) with ≥ 50 presence patches. After excluding 44 species with insufficient connectivity structure (zero mean current), 241 species (2 CR, 17 EN, 77 VU, 145 NT; 358 weight-units) contributed to the multi-species priority surface.

Protection baseline. Restricting the analysis to the 65,298 patches within Swiss territory, the current protected-area network covers 28.6 % of patches: 1.7 % under strict protection (IUCN categories Ia–III) and 21.4 % under managed protection (categories IV–VI), and an additional 5.5 % with unclassified IUCN category. The remaining 71.4 % (46,603 patches) lack any formal protection status. Note that this analysis considers a patch as protected if at least one of its constituent pixels overlaps with a protected area polygon—a generous criterion that may overstate actual coverage.

Gap analysis. The priority surface, combining threat-weighted suitability and connectivity, identifies patches that are simultaneously important habitat and critical corridor links for threatened species. Figure 6 shows the national-scale priority map overlaid with existing protected areas.

Among the top-5 % priority patches (3,265 patches):

- 58.9 % fall within some protected area (4.2 % strictly, 43.5 % managed, 11.2 % with unclassified IUCN category),
- **41.1 % (1,342 patches) are entirely unprotected:** these represent the primary conservation gaps.

Focusing on the top-1 % critical corridor patches (653 patches), protection coverage improves slightly: 6.4 % strict, 58.3 % managed, 9.5 % unclassified, but **25.7 % (168 patches) remain unprotected.** These 168 patches represent strong connectivity links between isolated populations of threatened species that currently have no formal protection.

4. Results and Discussion

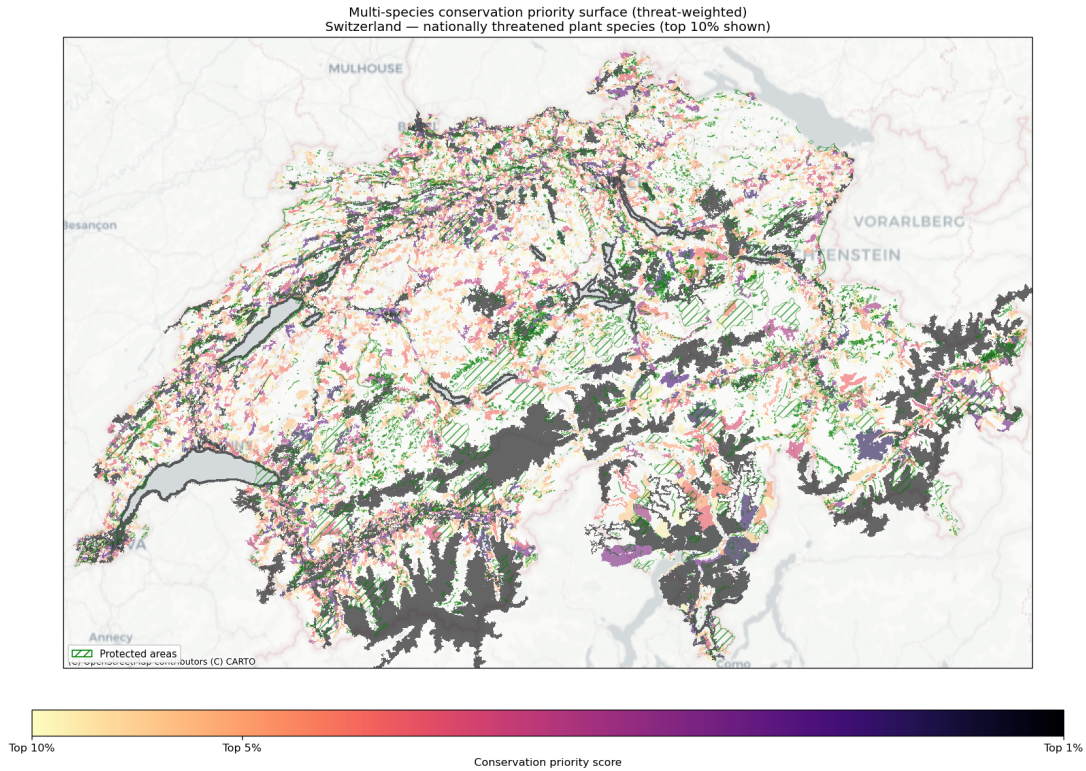


Figure 6: Multi-species conservation priority surface for Switzerland, aggregated across 241 nationally threatened species. High-priority patches (warm colours) combine high GNN-predicted habitat suitability with critical circuit-theory connectivity. Existing protected areas are outlined.

Regional patterns. The spatial distribution of conservation gaps is not uniform across Switzerland. Figure 7 illustrates four contrasting regions:

- **Engadin:** strong alignment between existing protected areas and priority patches: the extensive national park and surrounding reserves effectively cover critical connectivity corridors.
- **Seeland:** poor match: high-priority wetland and agricultural transition patches largely fall outside existing protection, despite their importance for threatened lowland flora.
- **Les Franches-Montagnes:** partial overlap: some priority grassland corridors are captured by landscape protection areas, but key connecting patches remain unprotected.
- **Surselva (west of Chur):** notable mismatch: several protected areas in this region do not coincide with high-priority patches, while critical corridors identified by the model lie outside existing designations.

The gap analysis reveals two categories of conservation action: (1) 168 unprotected critical corridor patches requiring new protection designation, and (2) 381 patches under managed protection (category IV–VI) that may warrant upgrading to stricter status given their connectivity importance. The finding that 41 % of high-priority habitat lacks any protection aligns with the broader assessment that Switzerland’s ecological infrastructure remains incomplete (Federal Office for the Environment, 2025).

The regional heterogeneity shows that protection effectiveness cannot be assessed by area alone. The Surselva example indicates protected areas optimised for landscape aesthetics or recreation do not necessarily capture the ecological corridors critical for threatened plant species.

4. Results and Discussion

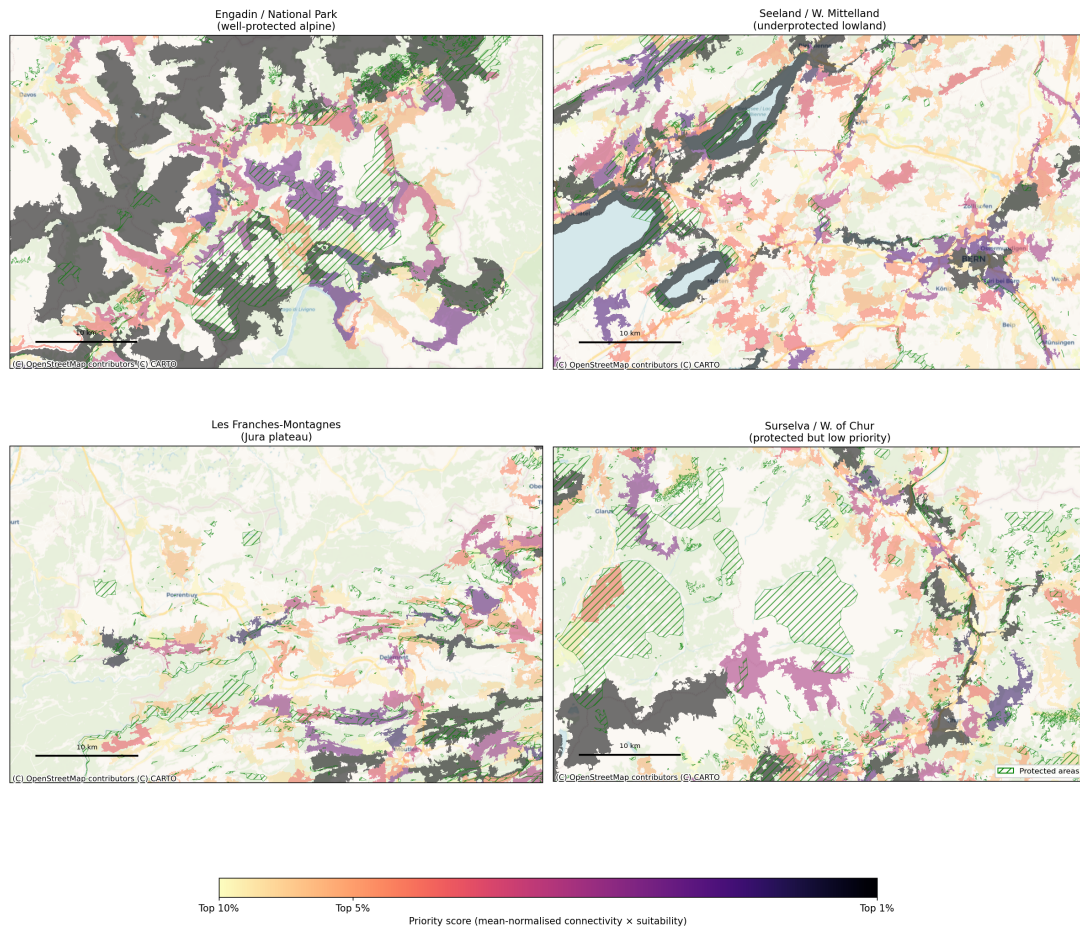


Figure 7: Regional detail of four contrasting areas showing the alignment (or mismatch) between the priority surface and existing protected areas. Engadin shows good coverage, while Seeland and Surselva reveal significant conservation gaps where high-priority habitat lacks protection.

5. Limitations and Future Work

Conversely, the Engadin demonstrates that well-placed protection can effectively cover connectivity needs.

5. Limitations and Future Work

5.1. Limitations

GBIF sampling bias. GBIF occurrence data is subject to strong spatial, temporal, and taxonomic biases. Observations cluster near roads, trails, and population centres; remote alpine areas and private land are systematically under-sampled. The model interprets absence from the database as pseudo-absence, which conflates true absence with lack of sampling effort. While this is standard practice in SDM and the overprediction of suitable habitat is a desirable fail-safe for conservation, it means predicted ranges should not be interpreted as precise distribution boundaries without field verification.

No independent field validation. Model performance is evaluated against held-out GBIF data, which shares the same sampling biases as the training set. Independent validation against systematically collected field surveys (e.g. Info Flora monitoring plots) would provide a stronger assessment of predictive accuracy, particularly for rare species in under-sampled regions.

Temporal snapshot. All environmental features and occurrence records represent a single temporal snapshot (climate normals 1981–2010, land cover 2021, GBIF records up to 2024). The model does not account for ongoing climate change, land-use intensification, or species range shifts. Priority areas identified today may lose relevance under future climate scenarios.

SOM patch size heterogeneity. The SOM-based discretisation produces patches that are adaptive in size, but the adaptation is imperfect. In homogeneous high-alpine terrain, patches can grow very large (>1000 pixels, ~ 0.9 km²), merging what may be ecologically distinct micro-habitats (e.g. snow-bed communities vs. wind-exposed ridges). Ideally, large alpine patches should be subdivided based on additional criteria such as aspect or micro-topography. This coarsening reduces spatial precision for alpine specialists.

Unweighted edges. The landscape graph uses binary adjacency: patches sharing at least one boundary pixel are connected with equal weight. This treats a single-pixel corridor under a road identically to a broad shared border between two forest patches. Edge weighting by shared-boundary length or by a resistance surface would better represent functional connectivity, particularly for the circuit-theory analysis where current should preferentially flow through wide corridors.

SOM quantisation loss. Assigning each 30 m pixel to its best-matching unit and then averaging features within patches introduces quantisation loss. Fine-scale environmental gradients within a patch (e.g. a south-facing rock outcrop within a north-facing forest patch) are smoothed away. This is the trade-off enabling graph-based modelling at national scale, but it limits predictions for species whose distributions are driven by micro-habitat variation.

6. Conclusion

Protected-area overlay simplification. The gap analysis treats protection as binary (any pixel overlap counts) and does not differentiate management effectiveness. A patch overlapping a category-IV landscape protection area receives the same “protected” label as one within a strict nature reserve, though actual conservation outcomes may differ substantially.

5.2. Future Directions

Climate scenario projections. Replacing static climate normals with CHELSA future projections (SSP scenarios) would enable forecasting of range shifts and identification of climate refugia. The pipeline architecture supports this directly: recomputing patch features under a future climate scenario and re-running the trained GNN yields projected suitability surfaces without retraining.

Edge-weighted graph. Replacing binary adjacency with resistance-based edge weights (derived from land-cover permeability or shared-boundary length) would improve the ecological realism of both the GNN message passing and the circuit-theory connectivity analysis.

Multi-species GNN. The current per-species training is computationally simple but cannot exploit correlations between species with similar habitat requirements. A joint multi-species architecture (e.g. species embeddings concatenated with patch features) could improve predictions for rare species by transferring information from ecologically similar common species.

Adaptive patch subdivision. Introducing a secondary subdivision step for large homogeneous patches (splitting by aspect, elevation band, or micro-topographic position) would improve spatial precision in alpine terrain without affecting the lowland patches that are already well-resolved.

Transfer to other regions. The pipeline is not specific to Switzerland: given environmental rasters and occurrence data, the full pipeline can be applied to any study area. Transfer to neighbouring Alpine countries (Austria, northern Italy) or to other taxa (insects, birds) would test the generalisability of the approach.

6. Conclusion

Graph neural networks can effectively model habitat suitability for Swiss flora at 30 m resolution. By encoding the landscape as a graph of 166,464 ecologically homogeneous patches linked by spatial adjacency, the SOM-GNN-SDM framework captures spatial context that traditional per-site models cannot represent.

Three main results:

1. **Graph structure improves predictions.** The GNN-SDM outperforms the Random Forest baseline for 85.3 % of 2,696 species (mean AUC 0.912 vs 0.874, mean TSS 0.583 vs 0.430), with the largest gains for species whose distributions depend on landscape connectivity rather than local features alone.

6. Conclusion

2. **The model generalises to rare species.** Nationally threatened species achieve comparable or higher discrimination (AUC 0.874, TSS 0.708) than common species, confirming that the framework is suitable for conservation applications despite lower observation density.
3. **Significant protection gaps exist.** Circuit-theory connectivity analysis across 241 threatened species reveals that 41 % of high-priority habitat patches—locations that are both suitable habitat and critical connectivity corridors—currently lack any formal protection.

These results should be interpreted as a methodological proof-of-concept rather than a direct policy recommendation. Conservation decisions involve many factors beyond ecology: a landscape patch identified as a critical ecological corridor may simultaneously be productive farmland, infrastructure, or subject to competing land-use interests. The priority surface identifies *where* ecological connectivity is most vulnerable, but the question of *whether* and *how* to act requires integration with socio-economic constraints, stakeholder consultation, and on-the-ground verification that lie beyond the scope of this analysis.

The full pipeline—from raw satellite and climate data to actionable conservation priority maps—runs end-to-end in under 10 hours of compute time and is readily transferable to other regions or taxa. As Switzerland continues to develop its ecological infrastructure strategy, data-driven approaches like SOM-GNN-SDM can complement expert assessment and field surveys with spatially explicit, reproducible evidence.

Acknowledgements

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This report does not constitute a policy recommendation of the FOEN. While the author is employed at FOEN and the programme was partially funded by FOEN, this work was carried out independently as an academic case study exploring the application of machine learning to conservation planning. Any interpretation of results in a policy context would require additional validation, stakeholder engagement, and integration with existing planning instruments.

AI assistance was used in the creation of this report as well as in the creation of the code used.

Code availability. The full analysis pipeline (notebooks 00–32) is available at <https://github.com/kronmar-bafu/cas-aml-project>.

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A. Supplementary Figures

Wu, Zhongyi et al. (2025). “GNN-SDM: A Graph Neural Network-Based Framework Integrating Complex Landscape Patterns Into Species Distribution Modelling”. In: *Global Ecology and Biogeography* 34, e70162. DOI: [10.1111/geb.70162](https://doi.org/10.1111/geb.70162).

A. Supplementary Figures

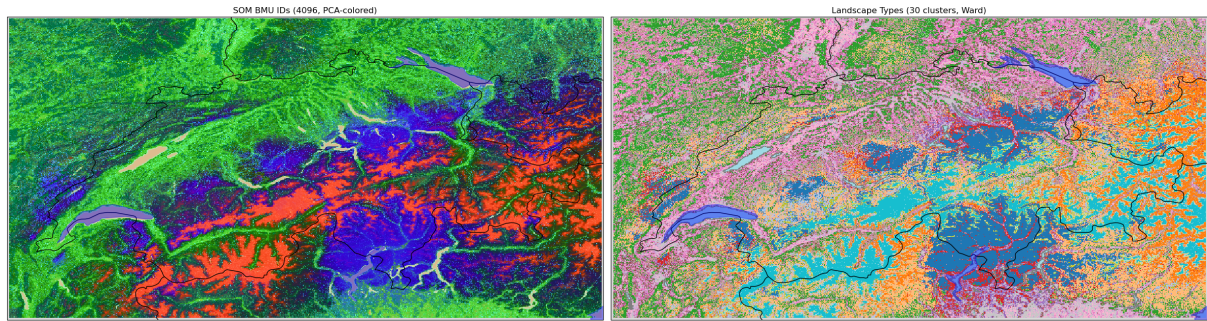


Figure 8: PCA-coloured landscape patch map of Switzerland. Codebook vectors are projected onto 3 principal components (mapped to RGB), so perceptually similar colours indicate environmentally similar patches. Major biogeographic zones are clearly distinguishable: lowland plateaus (Swiss Plateau, Po Valley) share warm tones, high-alpine areas appear as a distinct cool cluster, and forested pre-alpine slopes form intermediate gradients.

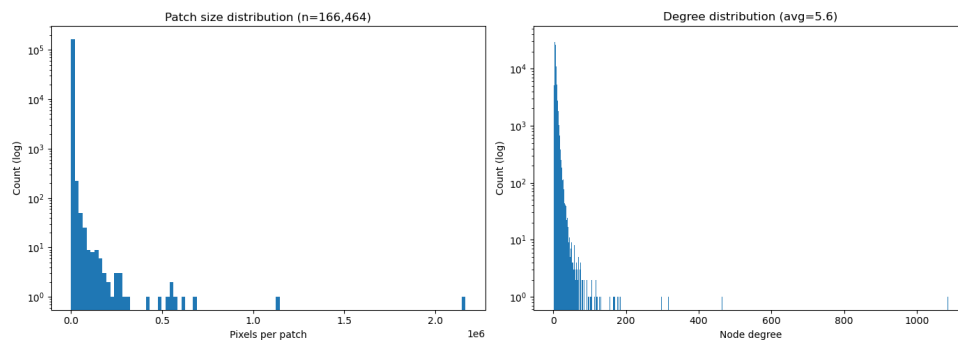


Figure 9: Patch size distribution (log scale) for the 166 464 landscape patches. The vertical dashed line marks the minimum threshold of 100 pixels ($\sim 0.09 \text{ km}^2$).

B. Random Forest Baseline Details

Table 4 summarises the cross-validated performance metrics for the Random Forest baseline across all 2,696 species.

The AUC distribution is strongly right-concentrated: 99.3 % of species achieve $\text{AUC} > 0.7$, and 91.6 % exceed 0.8.

B. Random Forest Baseline Details

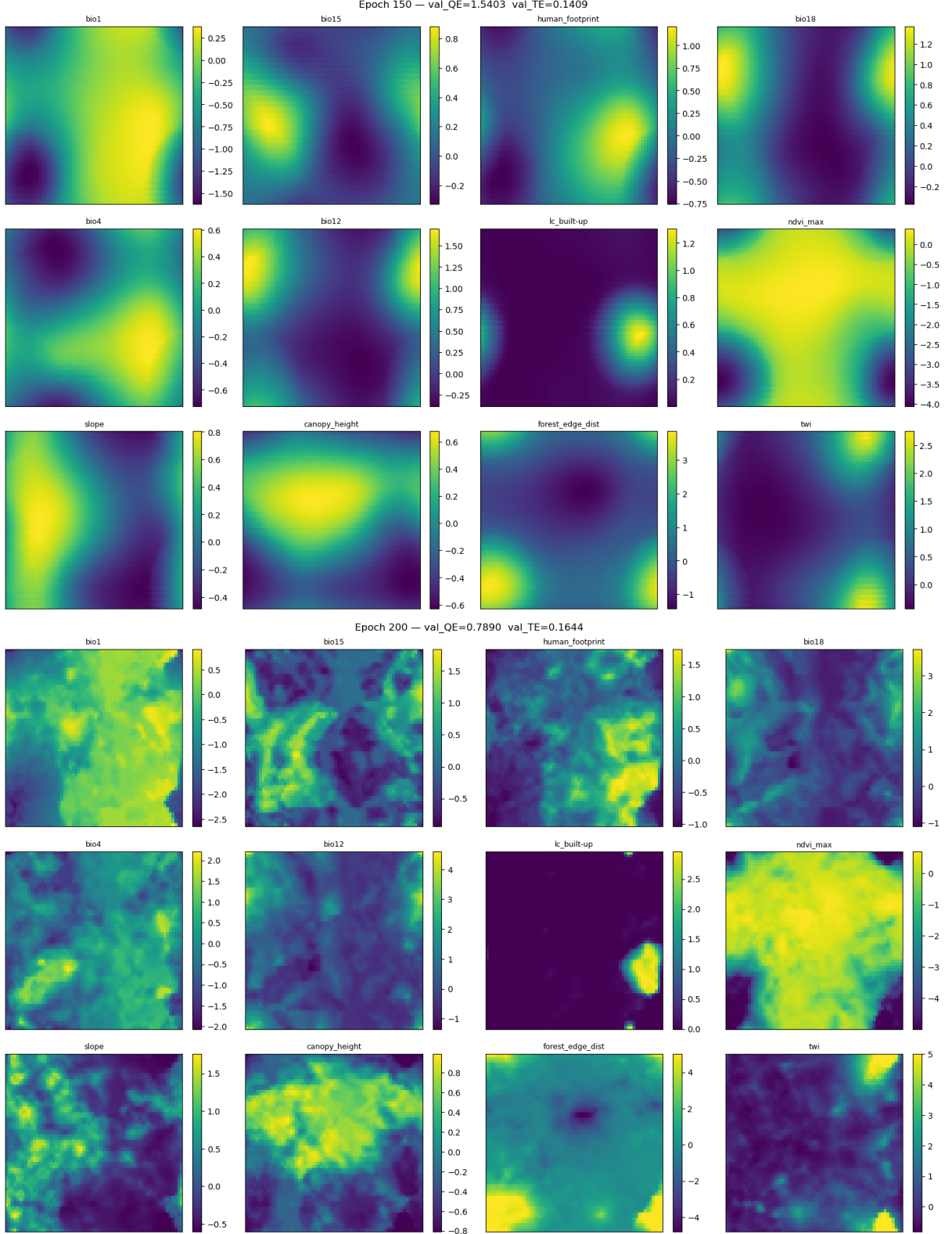


Figure 10: SOM codebook feature maps at epoch 150 (top) and epoch 200 (bottom). At 150 epochs the codebook shows smooth gradients with distinct regional structure across all 12 features. At 200 epochs, overtraining causes feature maps to fragment into noise—particularly visible in slope, TWI, and bio15—as the reduced neighbourhood radius allows individual neurons to chase local samples rather than preserving global topology.

C. GNN-SDM Training Diagnostics

Table 4: Random Forest baseline: summary statistics across 2,696 species (5-fold cross-validation, threshold = 0.5).

Metric	Mean	Median	Std
AUC	0.874	0.877	0.056
Accuracy	0.832	0.831	0.042
Precision	0.733	0.754	0.151
Recall	0.481	0.486	0.185
F1	0.561	0.585	0.173
MCC	0.498	0.512	0.163
TSS	0.430	0.433	0.176

C. GNN-SDM Training Diagnostics

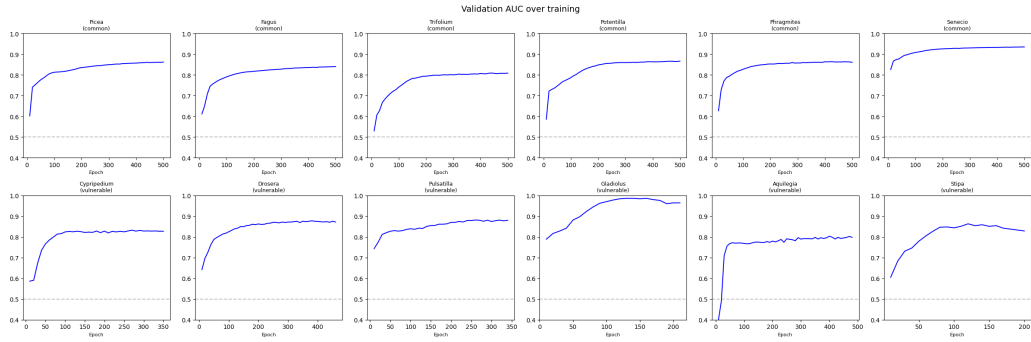


Figure 11: Validation AUC curves during GNN-SDM training for the 12 selected species (6 common, 6 vulnerable). Early stopping restores the model at peak validation AUC. Most species converge within 200–350 epochs; common species with more training data tend to use the full 500-epoch budget.

D. Per-Species Habitat Suitability Maps

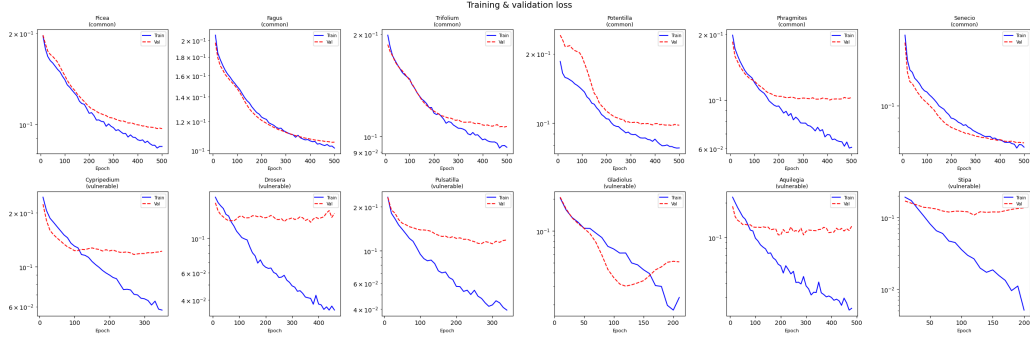


Figure 12: Training and validation loss (weighted MSE) for the 12 selected species. The gap between training and validation loss remains small throughout, indicating minimal overfitting. Dropout ($p = 0.2$) and early stopping provide sufficient regularisation.

D. Per-Species Habitat Suitability Maps

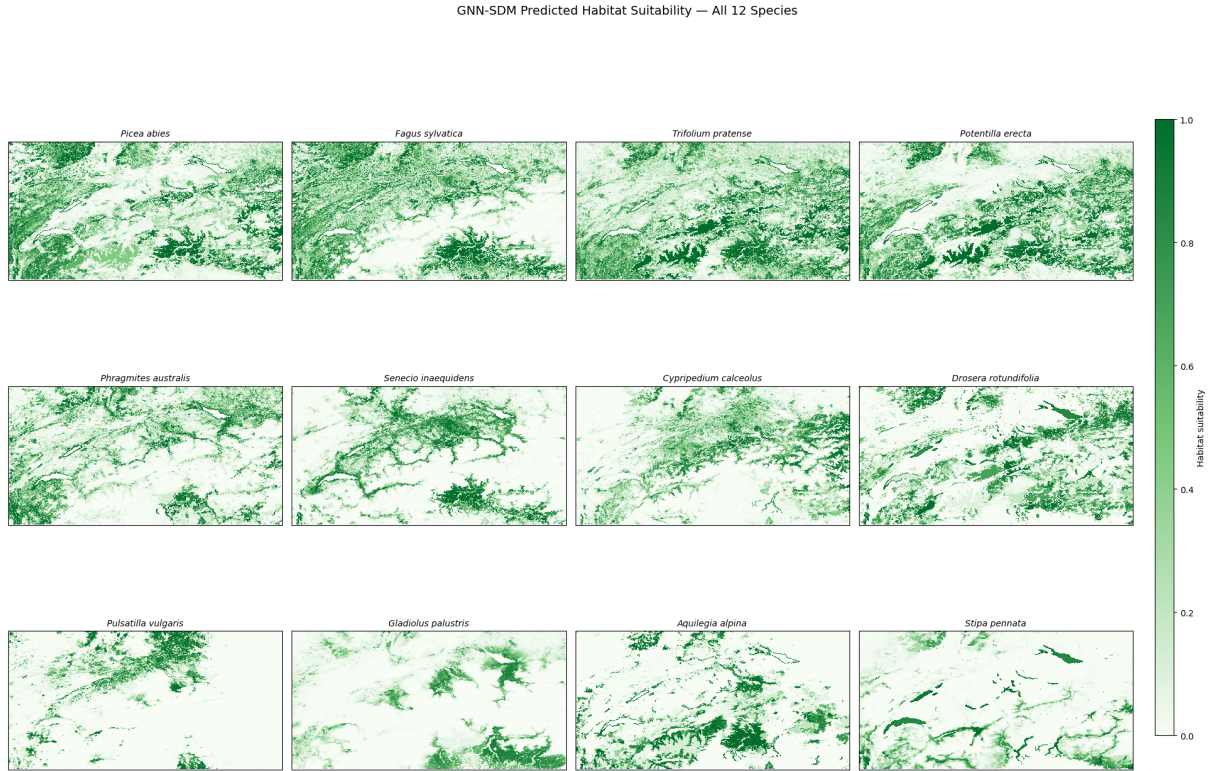


Figure 13: GNN-SDM habitat suitability maps for the 12 selected species. Warm colours indicate high predicted suitability. Common species (top two rows) show broader predicted ranges consistent with their generalist niches, while vulnerable species (bottom two rows) display concentrated suitability in restricted habitat patches.